

Elevator Pitch

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Tropical Medicine



Prescriptive Analytics

Optimization for Contract Design

Summary of Predictive Analytics

- From linear regression to neural networks
- Interpretability and accuracy
- Amount of data

Optimization Model: key elements



- Decision Variables
- Objective
- Constraints

Optimization

- Example: transportation problem
 - Four assembly plants, each requiring a number of engines shipped from various harbors

PLANT	REQUIRED
Leipzig	400
Nancy	900
Liège	200
Tilburg	500

HARBOR	AVAILABLE
Amsterdam	500
Antwerp	700
Le Havre	800

- Transportation costs

From Origin	To Destination			
	Leipzig	Nancy	Liège	Tilburg
Amsterdam	12	13	4	6
Antwerp	6	4	10	11
Le Havre	10	9	12	4

Transportation problem key elements

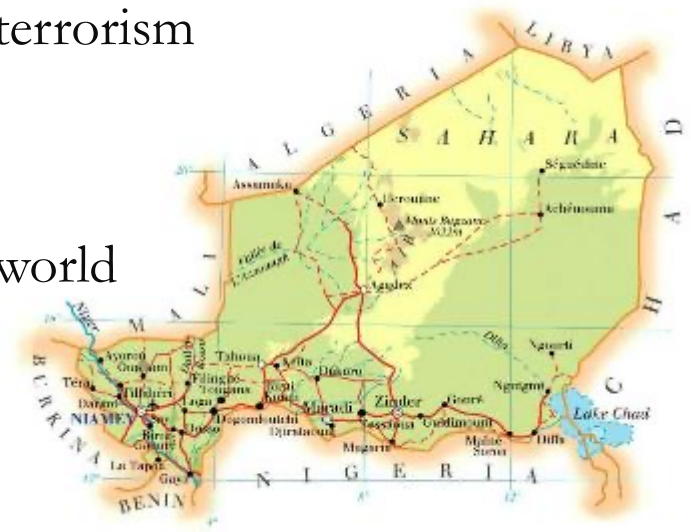
- Decision Variables
 - How much to ship from one origin to one destination (12 decision variables)
- Objective
 - Minimize total cost
- Constraints
 - Total available from each origin
 - Total required at each destination
 - Non-negativity

Model in Excel

	A	B	C	D	E	F	G
1	TRANSPORTATION PROBLEM						
2						Total cost	\$0.00
3		TRANSPORTATION COSTS					
4							
5		To Leipzig	To Nancy	To Liege	To Tilburg		No. available
6	From Amsterdam	12	13	4	6		500
7	From Antwerp	6	4	10	11		700
8	From LeHavre	10	9	12	4		800
9							
10	No. required	400	900	200	500		
11							
12	NUMBER TRANSPORTED						
13							Total
14	From Amsterdam						0
15	From Antwerp						0
16	From LeHavre						0
17							
18	Total	0	0	0	0		0

Fertilizer distribution in Niger

- Niger of West Africa, poor and flanked by terrorism
- A key: food insecurity
- One of the lowest users of fertilizer in the world
- Exceedingly high delivery costs
- Millennium Challenge Corporation (MCC)
- Risk of losing the agreement of \$437M to improve agriculture sector
- Endsight Consulting and the International Fertilizer Development Center (IFDC) to advise on fertilizer reform



Steve Grudda's initial model

Fertilizer Transport Optimization for Niger

IMPORTERS						
Transport Cost	IMP1	IMP2	IMP3	IMP4	IMP5	Available
Lome	2	3	2	1	2	15000
Coutonou	1	2	2	3	1	15000
Lagos	3	1	3	1	2	15000
Optimal Volume	IMP1	IMP2	IMP3	IMP4	IMP5	Total
Lome	-	-	3,000	6,000	-	9,000
Coutonou	6,000	-	3,000	-	6,000	15,000
Lagos	-	6,000	-	-	-	6,000
Total	6,000	6,000	6,000	6,000	6,000	30,000
Tender Approved	6000	6000	6000	6000	6000	30,000

Transport Cost to:	
Importers (5)	36,000
Regional Center (8)	20,000
Departments (36)	
Communes (265)	
Cantons/Groupement	
Village (17,000)	
Total Cost (Minimized)	56,000

REGIONAL CENTERS									
Transport Cost	Niamey	Dosso	Diffa	Zinder	Tilaberi	Agadez	Maradi	Tahoua	Available
IMP1	4	1	3	2	2	2	1	4	6,000
IMP2	3	4	3	2	2	4	1	2	6,000
IMP3	3	2	4	3	4	1	1	1	6,000
IMP4	1	3	2	3	1	2	3	1	6,000
IMP5	1	1	1	1	4	3	3	4	6,000
Optimal Volume	Niamey	Dosso	Diffa	Zinder	Tilaberi	Agadez	Maradi	Tahoua	Total
IMP1	-	1,500	-	-	-	-	500	-	2,000
IMP2	-	-	-	-	-	-	-	-	-
IMP3	-	-	-	-	-	2,500	2,000	1,500	6,000
IMP4	2,500	-	-	-	2,500	-	-	1,000	6,000
IMP5	-	1,000	2,500	2,500	-	-	-	-	6,000
Total	2,500	2,500	2,500	2,500	2,500	2,500	2,500	2,500	20,000
#USED AT RCs	2500	2500	2500	2500	2500	2500	2500	2500	20,000

Optimization for Contract Design

Incentives for Drug Development

- Nature of drug development
 - Clinic trials as a “gold standard”
 - Long time
 - High cost
 - Uncertain outcome
 - High drug prices to recoup investment
 - Drug development is about not only science, but also economics
 - Lack of profitable market for rare and tropical disease drugs
- ↓
- Lack of available drugs for rare and tropical diseases
 - Funding from governments and private foundations



<https://www.henryford.com/blog/2023/04/what-happens-during-each-phase-of-a-clinical-trial>

Global Health Funding

- Gates Foundation
 - Launched in 2000
 - \$77B assets as of late 2024
 - New deadline (May 2025):
 - “20 years to give away all (\$200B, including future contributions)”
 - Global Health Division
 - Has donated more than \$6.6 billion for global health programs
- The Global Fund
 - The Global Fund to Fight AIDS, Tuberculosis and Malaria
- Accountability

Based on part of this paper

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Optimal Push, Pull, and Failure Funding for Global Health

64 Pages • Posted: 10 Jun 2025

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Date Written: May 23, 2025

Abstract

Malaria caused over 600,000 deaths in 2021, yet commercial incentives are weak for drug and vaccine development for malaria and other tropical diseases. Governments and nonprofits address these market failures through push (e.g., grants) and pull (e.g., prizes). We propose a third mechanism in which the funder only pays when the firm fails. That is, the funder reimburses the firm a share of its testing costs, which we refer to as "insurance." Insurance is optimal when the commercial market is large enough to reward success, but too small to induce investment. The optimal mechanism addresses adverse selection and moral hazard. For most tropical diseases, including malaria, we recommend pull funding with supplementary push support. Insurance is optimal for tuberculosis if testing costs are less than a billion dollars. These findings challenge current practices dominated by push funding and extend to funding innovations in other sectors.

Keywords: mechanism design, adverse selection, moral hazard, global health, innovation

Funding Drug Development

- A firm with a patent for treating a tropical disease
- A foundation provides grants to support clinical trials
- Firm's private information
 - Firm knows how promising the drug is – Adverse selection
 - Firm needs to exert effort to maximize chance of success – Moral hazard
- Contract to induce firm to **truthfully reveal** the drug's promise; **exert effort** to maximize chance of success; and do it with the least cost

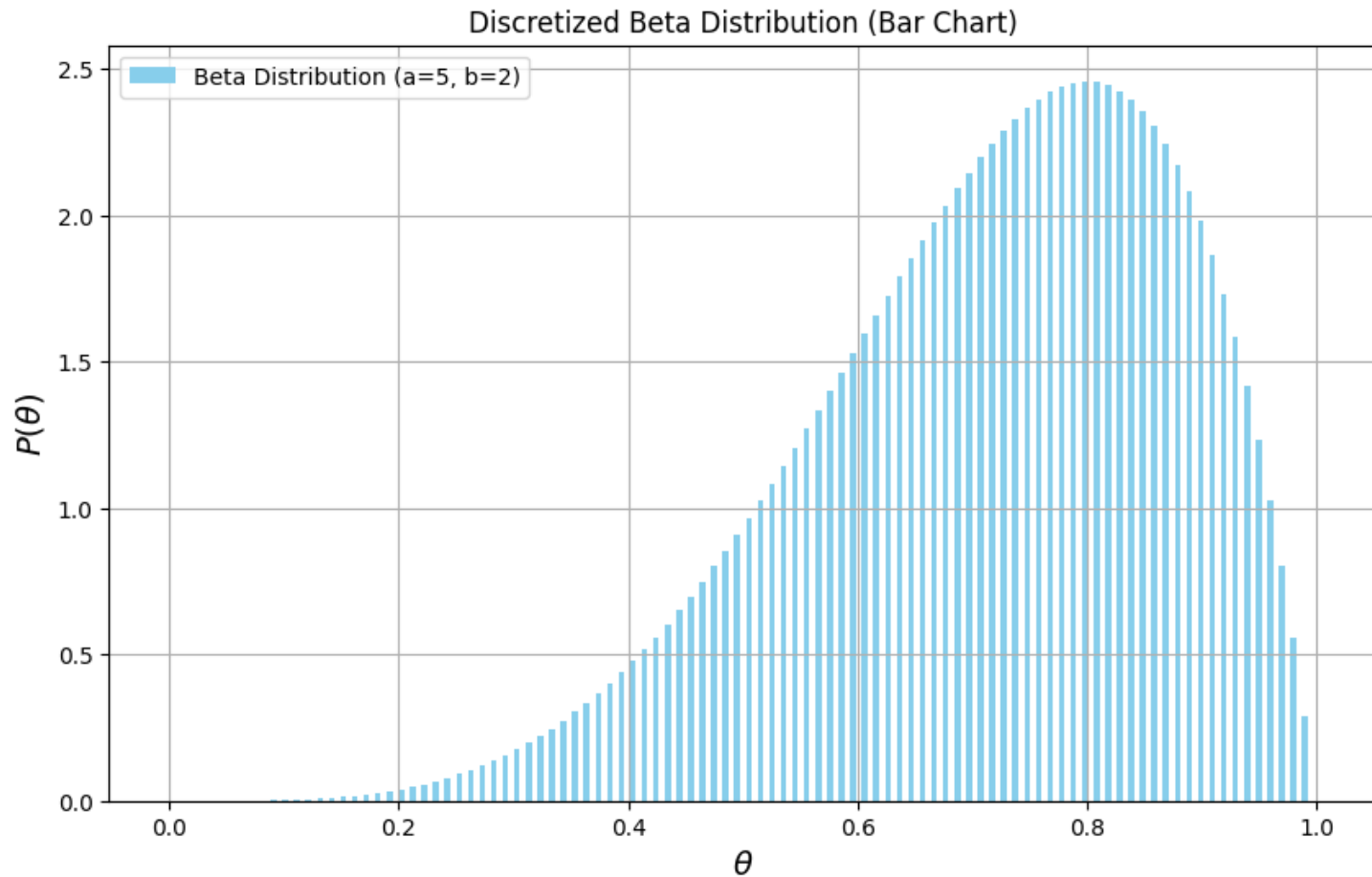
Push vs Pull Funding

- Push: offer funding in the beginning
 - Preferred by cash-constrained firm
 - Firm discounts future income with a factor $\delta < 1$ (say 0.6, for income a few years after clinical trial finishes)
 - Foundation does not discount (deeper pocket)
- Pull: offer funding after success (FDA approval)
 - Accountability
 - More costly – need to pay interest due to firm's discounting
- What about funding after failure?
 - Add the decision variable and see if optimization picks it up

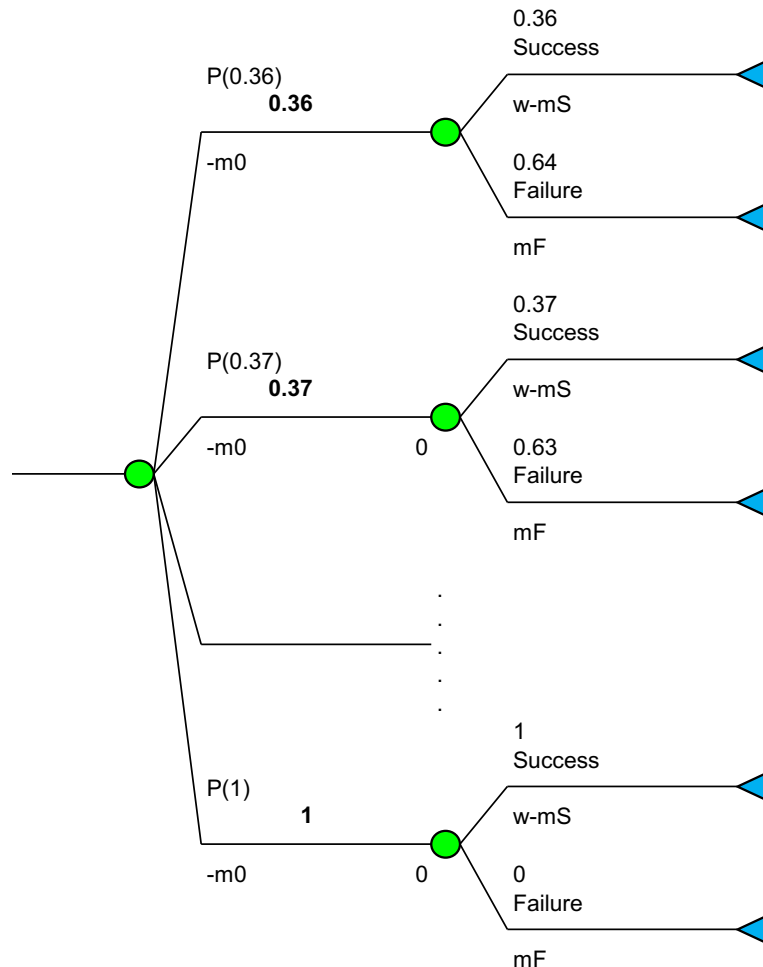
Model Parameters

- If drug is successful
 - Gates Foundation receives social **welfare** benefit: w (vary greatly across diseases)
 - Firm receives commercial **profit**: π (also vary greatly – whether high-income countries also suffer from the disease)
- Firm's effort cost: c (say, \$100M)
- If firm pretends to have worked, cost is $\tilde{c} < c$ and chance of success is 0. (say, $\tilde{c} \sim 0.5c$)
- With effort, the drug succeeds with probability θ
 - Firm knows θ
 - Foundation does not know θ , only knows distribution of θ

Distribution $P(\theta)$



Objective



- Foundation's probability on the probability of success being θ : $P(\theta)$
- m_0 : push (how much to pay **before** clinical trial)
- m_S : pull (how much to pay **after** clinical trial **succeeds**)
- m_F : “insurance” (how much to pay **after** clinical trial **fails**)

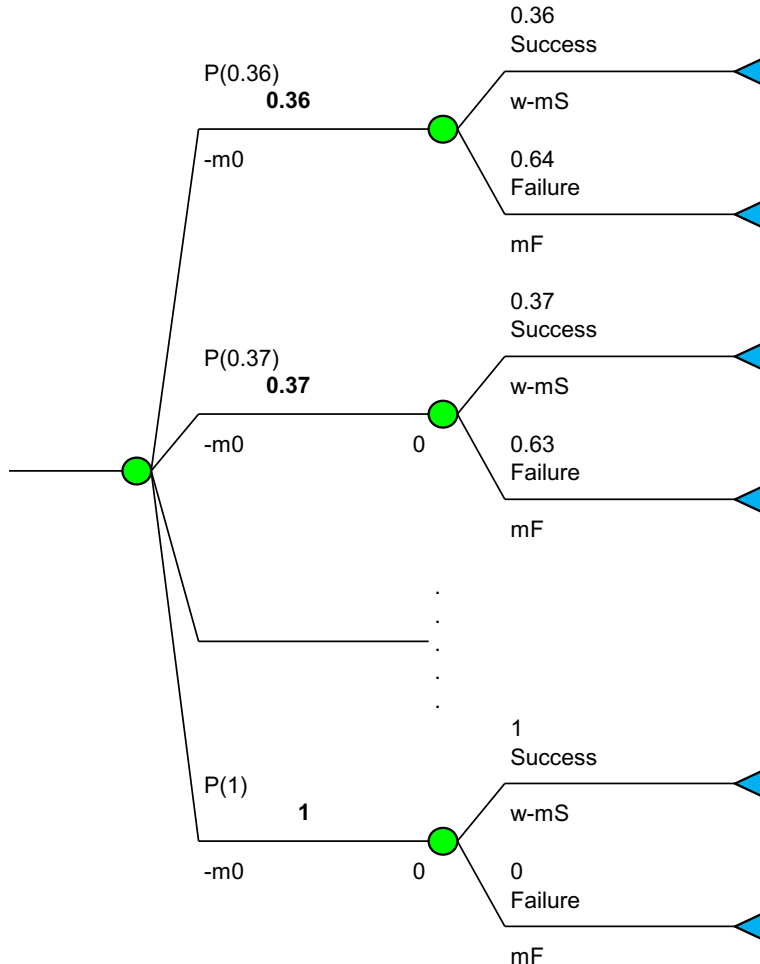
Optimal Contract Design

- **Decision variables** (3 of them):
 - m_0 : push (how much to pay **before** clinical trial)
 - m_S : pull (how much to pay **after** clinical trial **succeeds**)
 - m_F : “insurance” (how much to pay **after** clinical trial **fails**)
- These three funding levels imply a threshold, τ , on the chance of success, above which the firm is willing to participate
- **Objective:** Gates maximizes its expected payoff

$$\max -m_0 + \sum_{\theta \geq \tau} [\theta(w - m_S) - (1 - \theta)m_F] \cdot P(\theta)$$

- No discounting – deep pocket

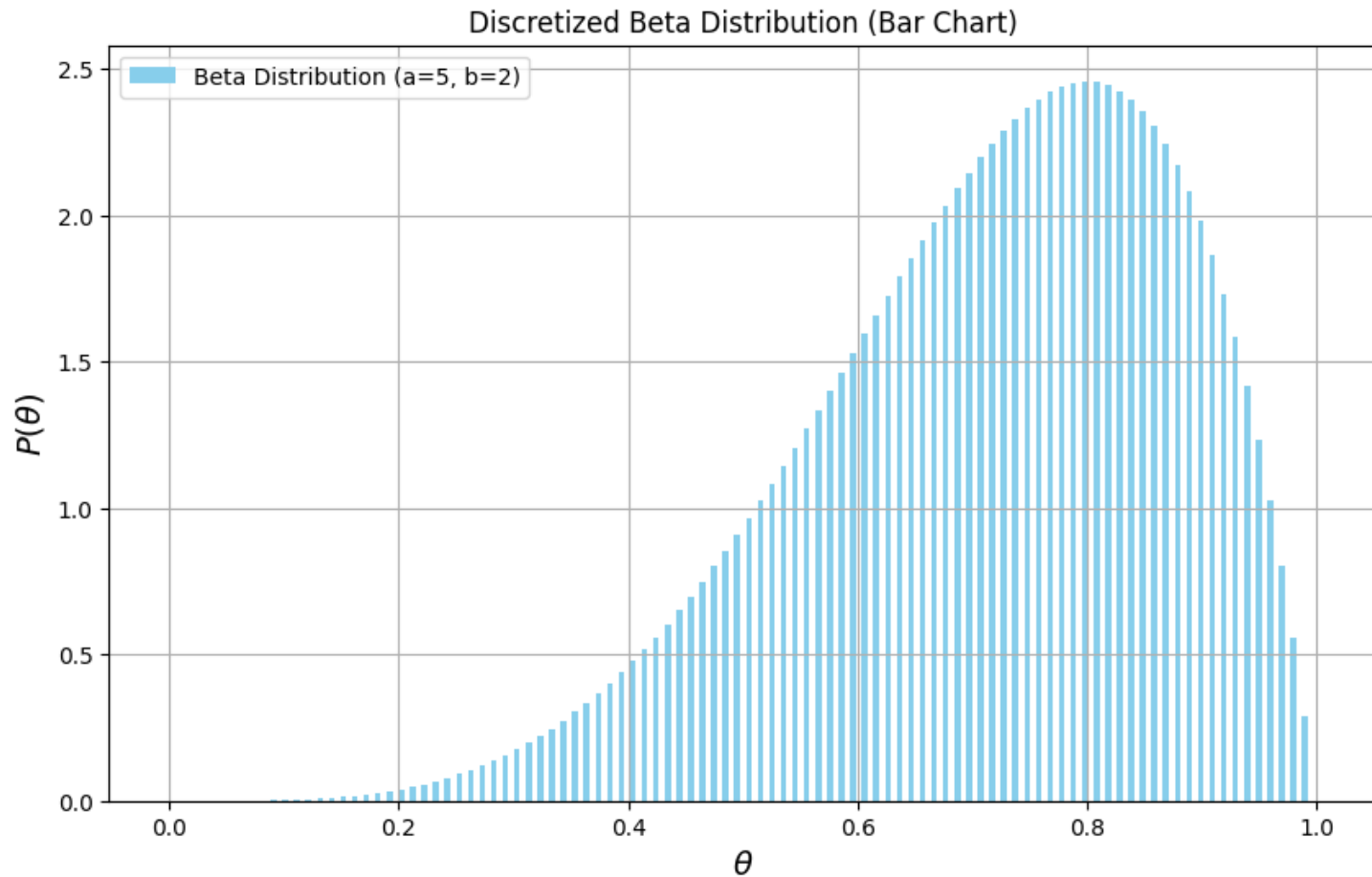
Objective function



Assume the firm is willing to participate if it believes that the probability of success is above $\tau = 0.35$.

$$\begin{aligned}
 & -m_0 \\
 & + P(0.36)[0.36(w - m_S) + 0.64(-m_F)] \\
 & + P(0.37)[0.37(w - m_S) + 0.63(-m_F)] \\
 & + P(0.38)[0.38(w - m_S) + 0.62(-m_F)] \\
 & + \dots \\
 & + P(0.99)[0.99(w - m_S) + 0.01(-m_F)] \\
 & + P(1)(w - m_S)
 \end{aligned}$$

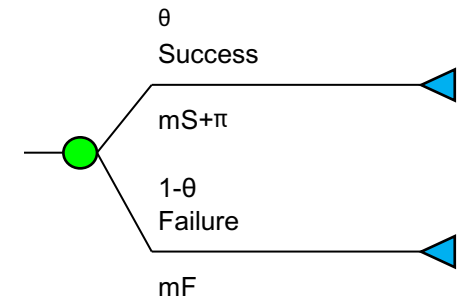
Distribution $P(\theta)$



Constraints

- Firm's expected gain if exerting effort:
 - From before clinical trial: $m_0 - c$
 - From clinical trial:

$$\delta[\theta(m_S + \pi) + (1 - \theta)m_F]$$



- Together:

$$(m_0 - c) + \delta[\theta(m_S + \pi) + (1 - \theta)m_F]$$

- Not exerting effort:

$$(m_0 - \tilde{c}) + \delta m_F$$

Constraints

- Firm's expected gain if exerting effort

$$\begin{aligned} & (m_0 - c) + \delta[\theta(m_S + \pi) + (1 - \theta)m_F] \\ &= (m_0 - c) + \delta\theta(m_S + \pi - m_F) + \delta m_F \end{aligned}$$

- Willing to exert effort if and only if chance of success $\theta \geq \tau$
 $(m_0 - c) + \delta\theta(m_S + \pi - m_F) + \delta m_F \geq (m_0 - \tilde{c}) + \delta m_F, \quad \forall \theta \geq \tau$

and

$$(m_0 - c) + \delta\theta(m_S + \pi - m_F) + \delta m_F < (m_0 - \tilde{c}) + \delta m_F, \quad \forall \theta < \tau$$

(many constraints)

- Monetary payments are non-negative: $m_0, m_S, m_F \geq 0$

Simplify Constraints

- Firm's utility of exerting effort to conduct clinical trial

$$\begin{aligned} & (m_0 - c) + \delta[\theta(m_S + \pi) + (1 - \theta)m_F] \\ &= (m_0 - c) + \delta\theta(m_S + \pi - m_F) + \delta m_F \end{aligned}$$

- Willing to exert effort if and only if chance of success $\theta \geq \tau$
 $\delta\tau(m_S + \pi - m_F) \geq c - \tilde{c},$

For θ *right below* threshold τ :

$$\delta\theta(m_S + \pi - m_F) \leq c - \tilde{c}$$

(2 constraints)

- Monetary payments are non-negative: $m_0, m_S, m_F \geq 0$

Optimization for Contract Design

$$\max \sum_{\theta \geq \tau} [-m_0 + \theta(w - m_S) - (1 - \theta)m_F] \cdot P(\theta)$$

Subject to

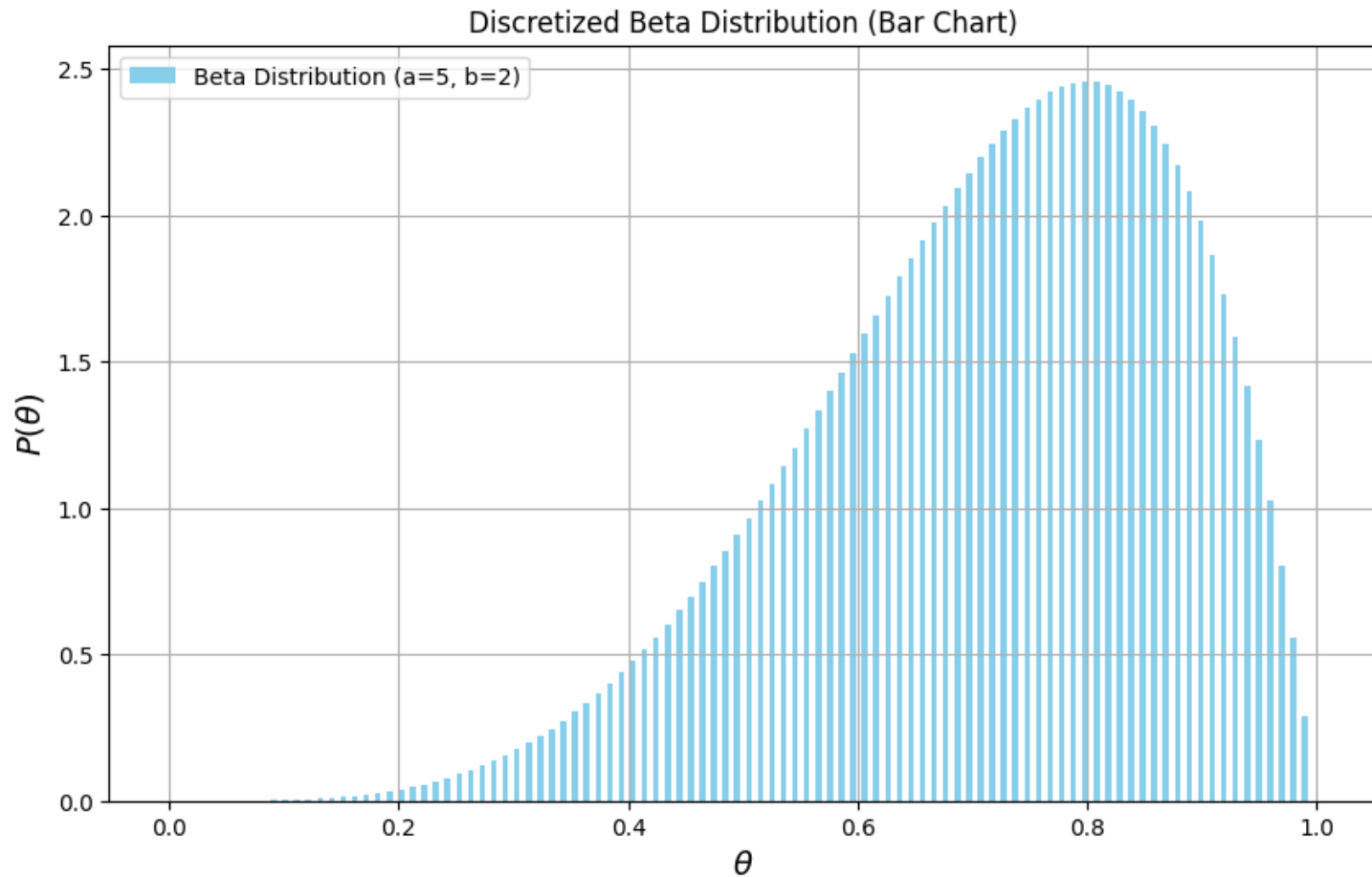
$$\delta\tau(m_S + \pi - m_F) \geq c - \tilde{c}$$

$$\delta\hat{\theta}(m_S + \pi - m_F) \leq c - \tilde{c}$$

$$m_0, m_S, m_F \geq 0$$

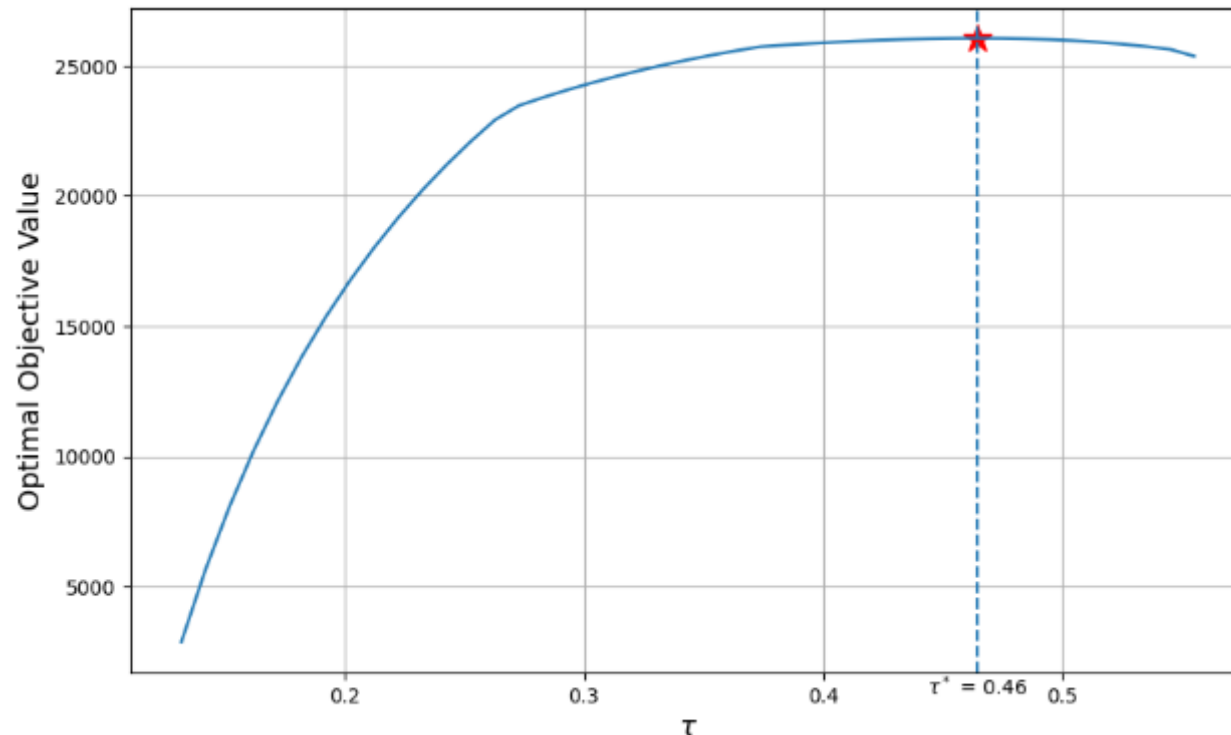
- For any given threshold τ , this is a linear optimization with 3 decision variables and 5 constraints.
- To find the best threshold τ , loop through possible values of θ !

Distribution $P(\theta)$



Optimize over different τ

- Fix $\delta = 0.6$, $c = \$100M$, $w = \$400M$, $\pi = \$300M$, $\tilde{c} = \$50M$



- Optimal contract: $m_0^* = 0$, $m_S^* = 0$, $m_F^* = \$46.15M$, $OBJ^* = \$26.06M$

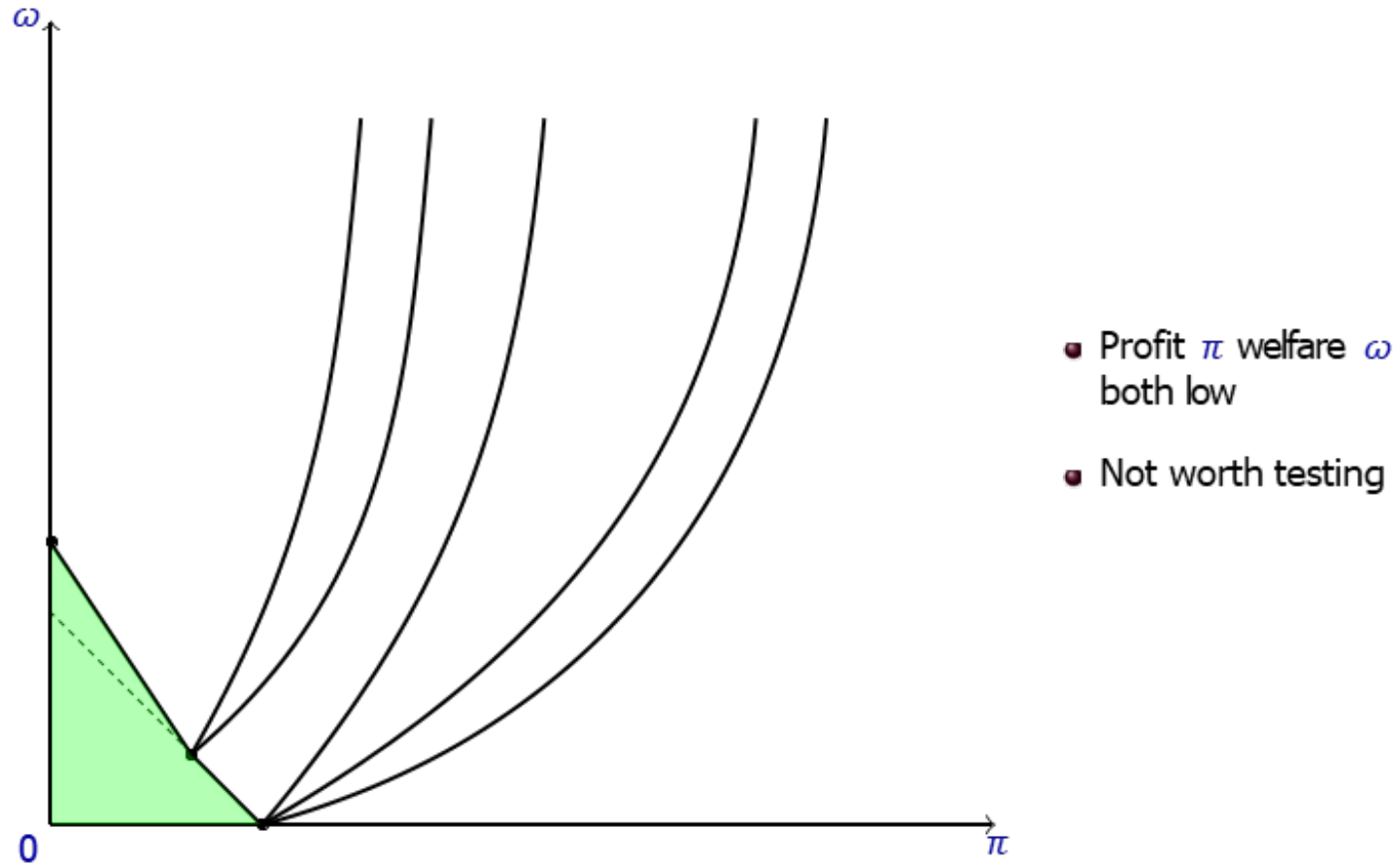
Economic Insight: why funding failure?

- What if we do not allow funding failure?
 - How to revise the optimization model to reflect this restriction?
 - Add constraint $m_F = 0$ (equivalent to removing this decision variable)
 - $OBJ^* = \$25.88M$, $\tau^* = 0.49$
- Why funding for failure helps?
 - Incentivize not-so-promising drug a chance to be developed
 - Offer “insurance”
 - Mitigate incentive to undersell (“sand-bag”)
 - “Give me a lot of push/pull because the drug will likely fail”
- Never optimal to pay for both success and failure
 - Better to increase push, and reduce pull and insurance, until one of them becomes 0

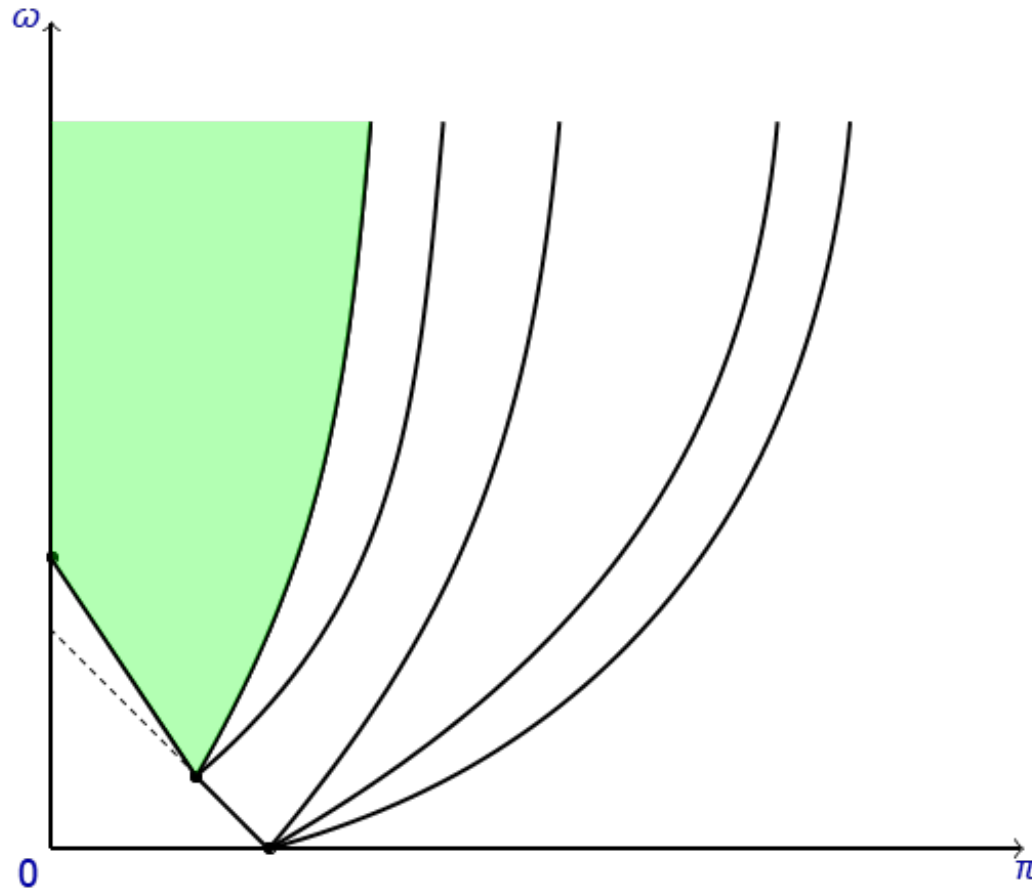
Overall picture

- Optimal solutions for varying model parameters
- Assume Uniform[0,1] distribution of θ
- Paper proves the following intuitive structures...

Case 1: $m_0^* = m_S^* = m_F^* = 0$, $\tau^* = 1$

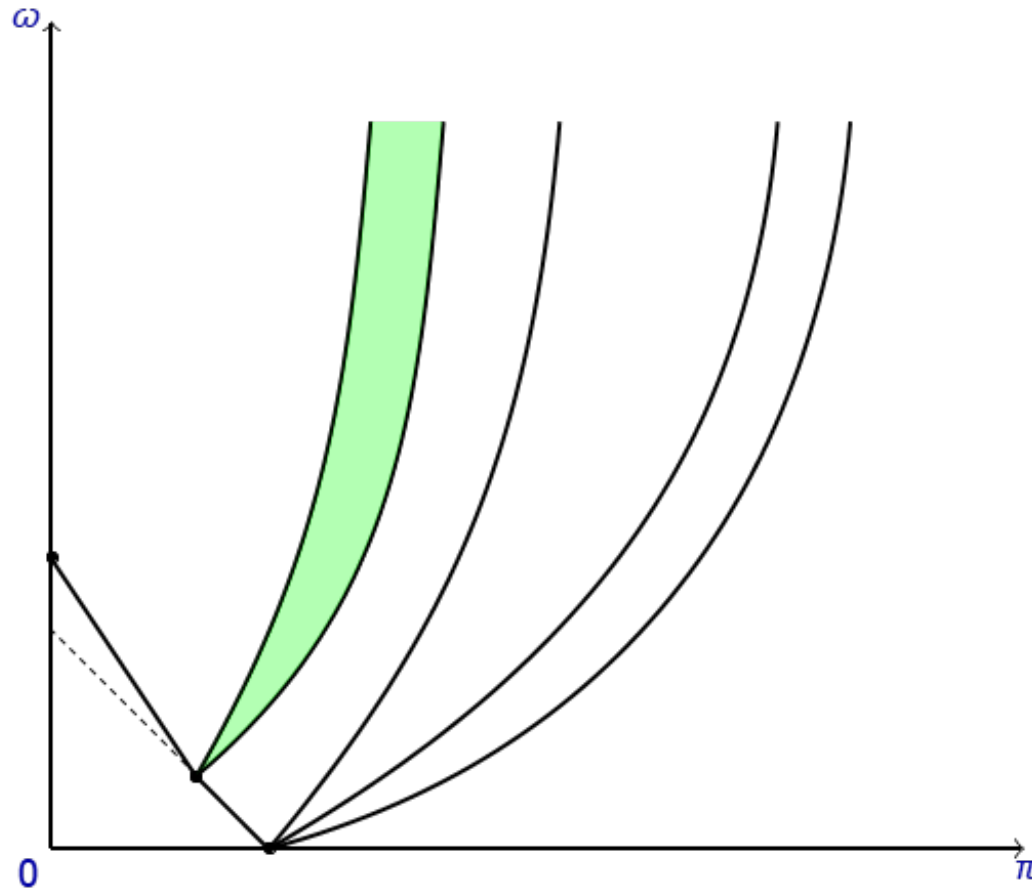


Case 2: $m_0^* = \tilde{c}$, $m_S^* > 0$, $m_F^* = 0$



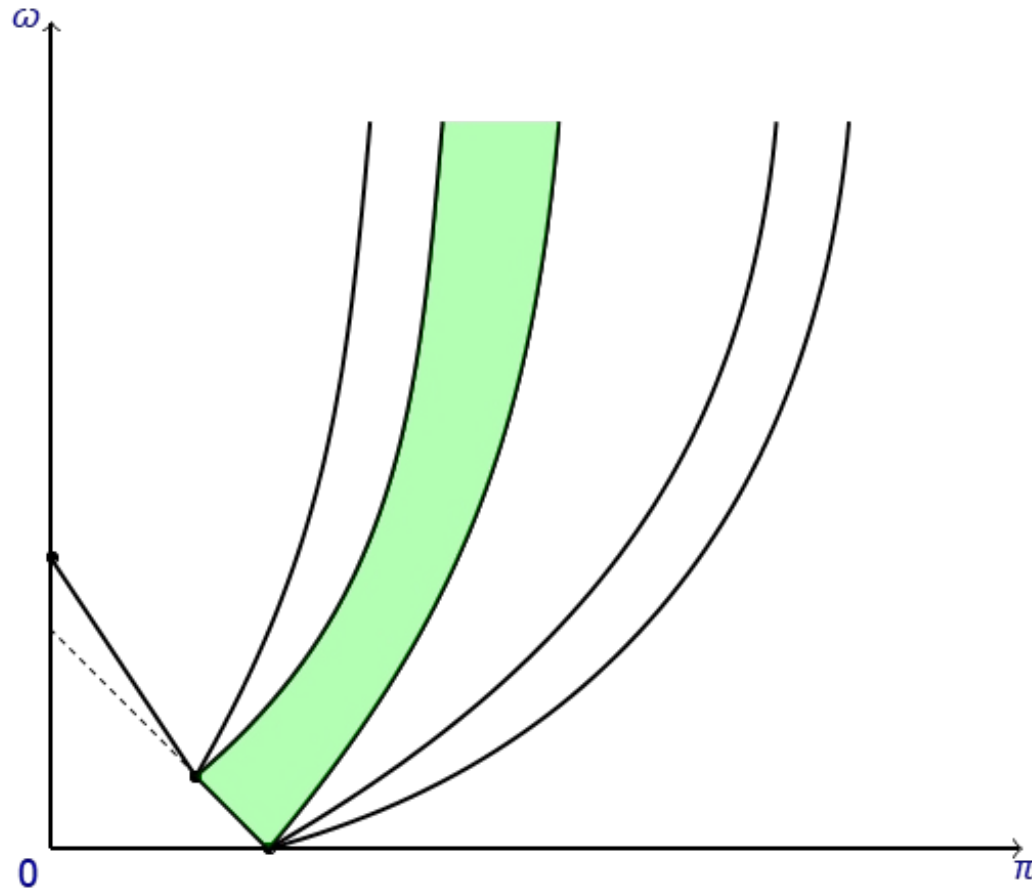
- Profit π low; welfare ω high
- Push (m_0) + pull (m_S)
- Intuitive

Case 3: $m_0^* = \tilde{c}$, $m_S^* = m_F^* = 0$



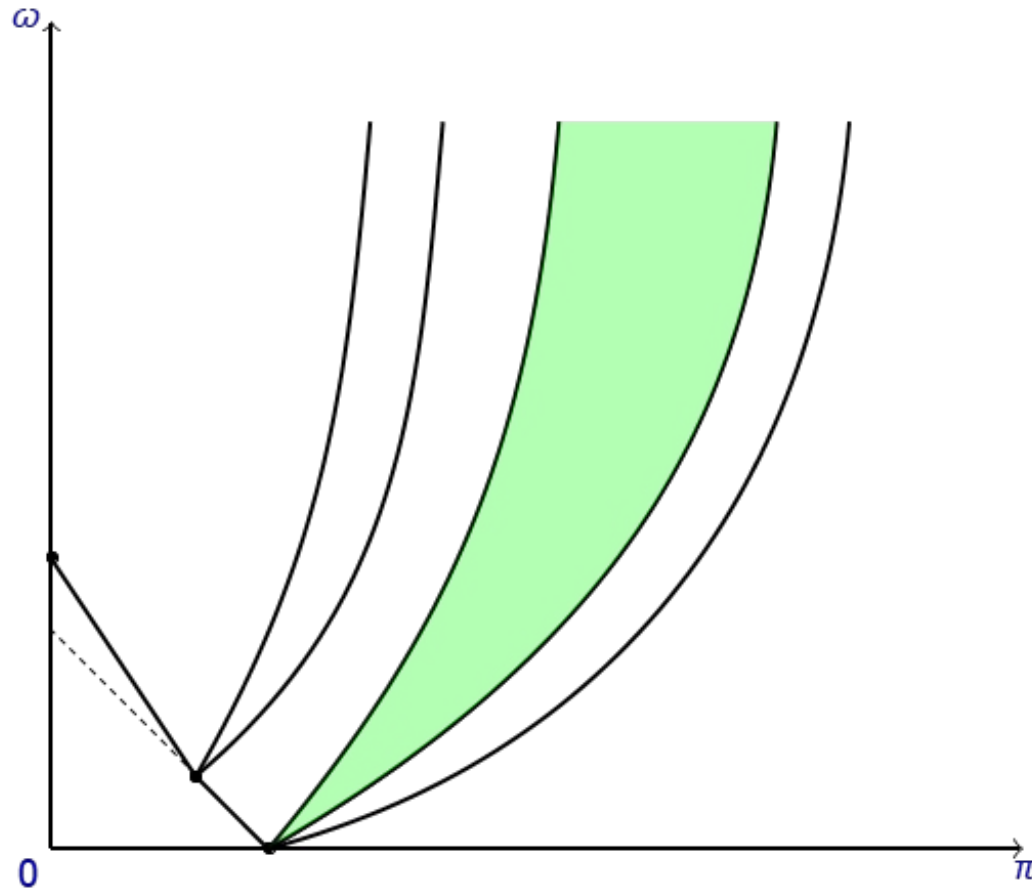
- Profit π sufficiently high
- No need to reward success – push only
- Keep m_0 at $\tilde{c}$:

Case 4: $m_0^* + \delta m_F^* = \tilde{c}$, $m_S^* = 0$



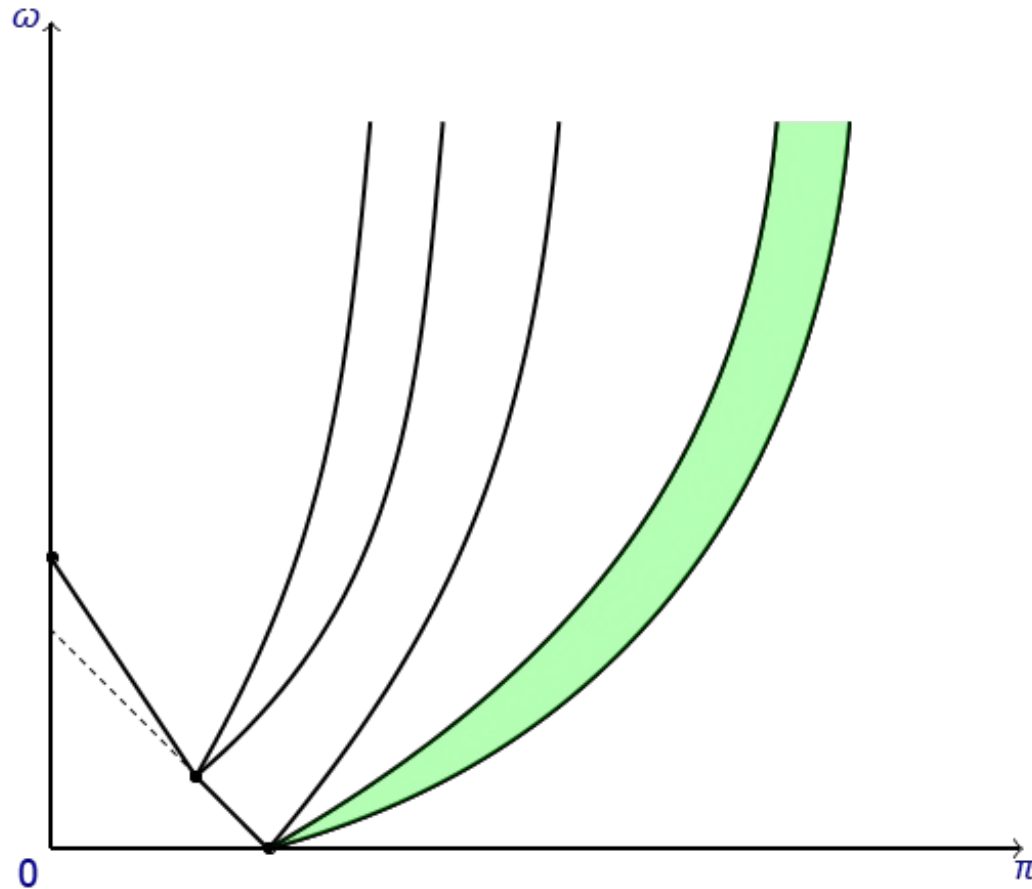
- Even higher profit π
- Rewarding failure!
- Sufficient reward for success from π
- Insurance m_F^* for failure
- Mitigate incentive to under-report θ
- Trade off due to discounting

Case 5: $m_0^* = m_S^* = 0, \delta m_F^* = \tilde{c}$



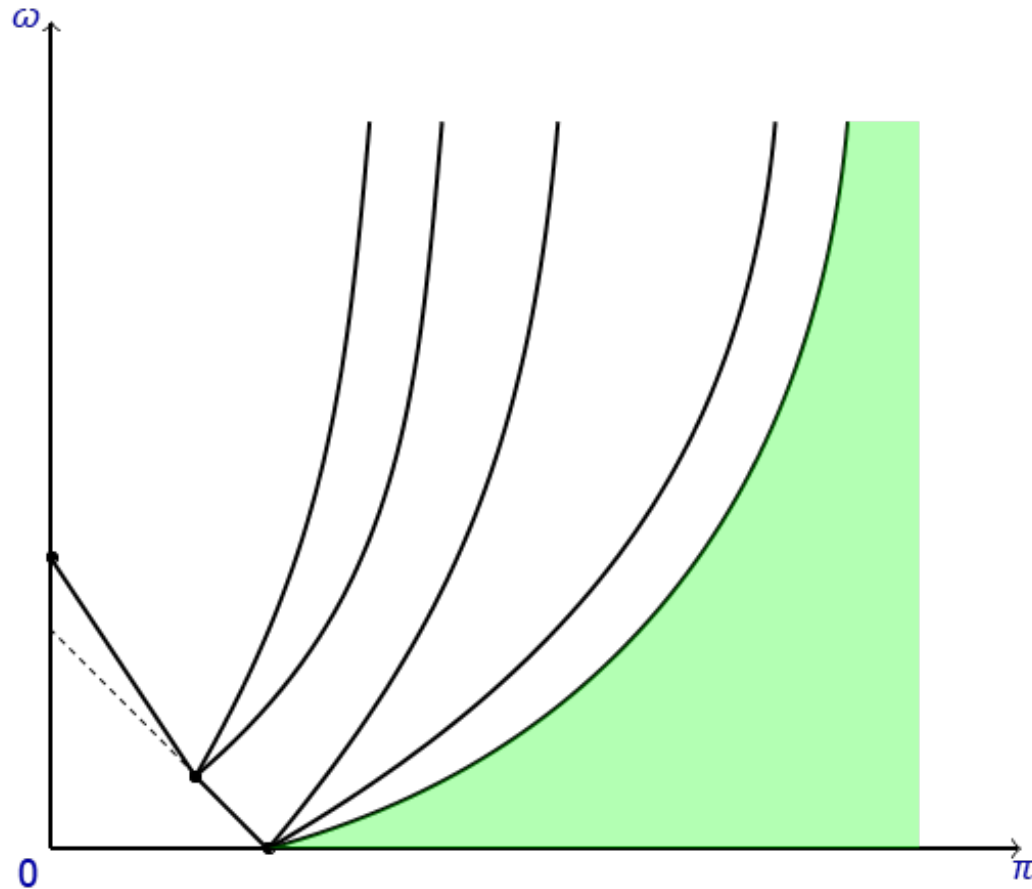
- Further increase π
- Only insurance (m_F),
no push (m_0)
- Present value αc
- τ decreases in π

Case 6: $m_0^* = m_S^* = 0, \delta m_F^* < \tilde{c}$



- Profit π higher – no need to keep max insurance level
- τ may increase in π – trade-off between failure reward and supporting low-chance drug

Case 7: $m_0^* = m_S^* = m_F^* = 0$, $\tau^* < 1$



- Profit π high enough for commercial market to take over
- Threshold τ keeps decreasing in π

Implementations

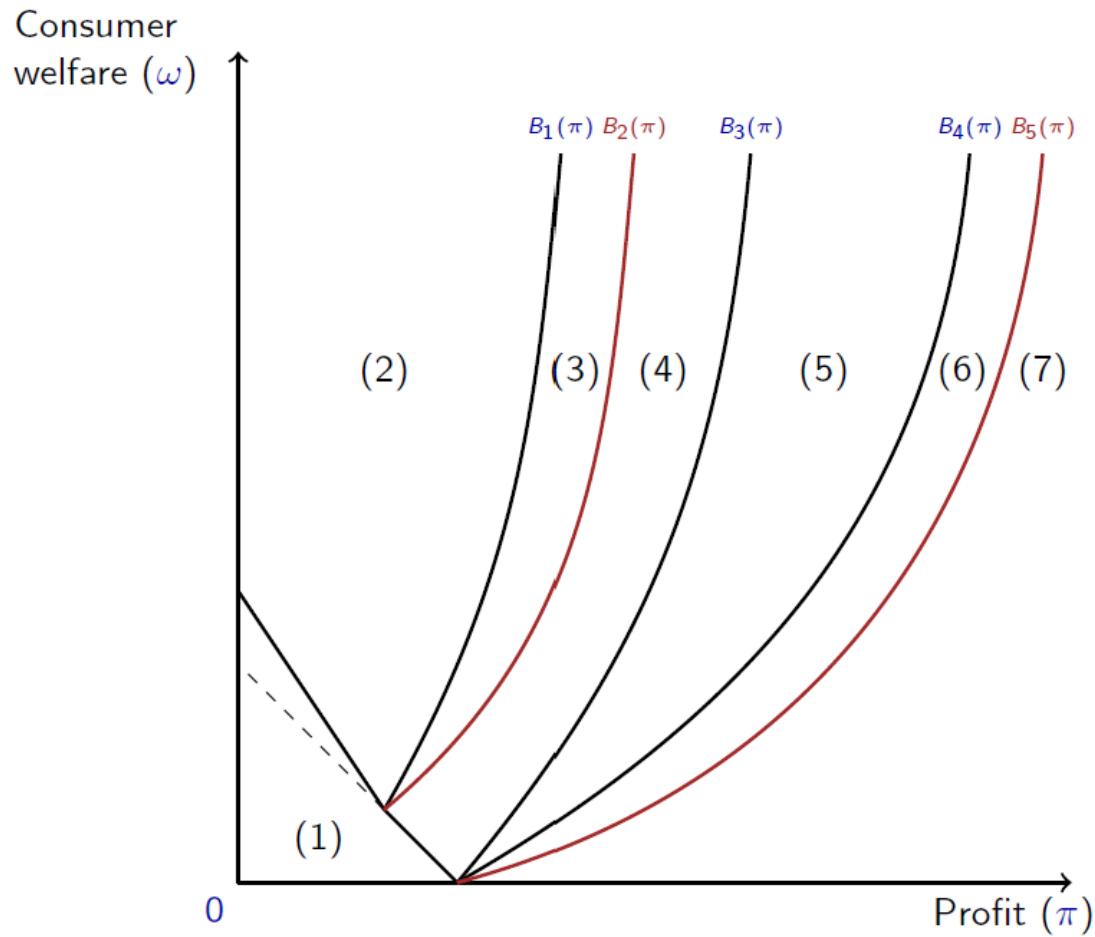
- How to calibrate the model parameters
 - Profit π : high-income countries' DALY (disability-adjusted life year) improvement
 - Welfare w : DALY improvement from all countries
 - Discount δ : commercial loan annual interest rate compounded over 3 to 4 years
- How to implement $m_F^* > 0$?
 - Reimbursement?
 - Forgivable loan?

Tropical Diseases

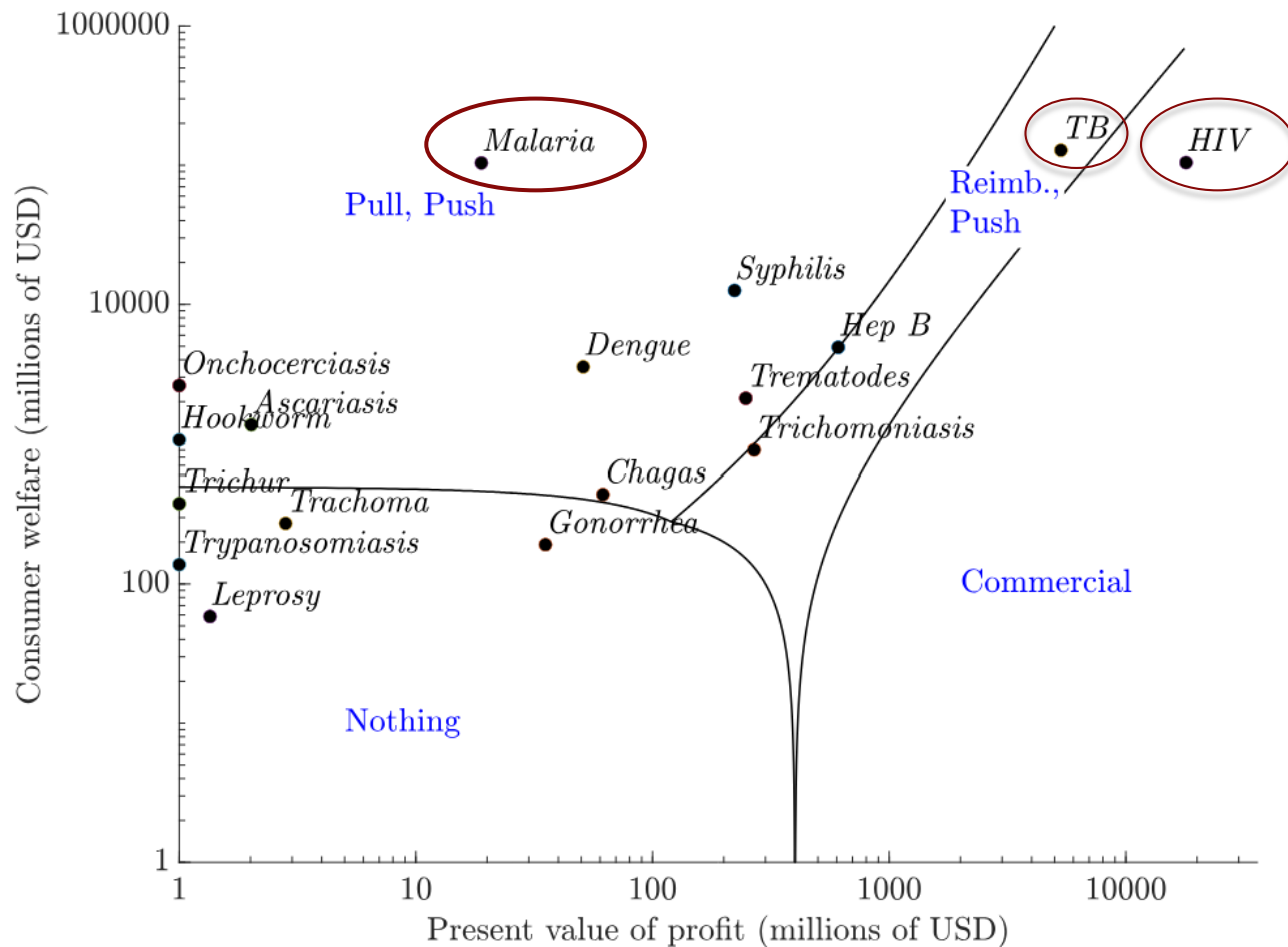
Disease	Type	Neglected	Annual DALYs (000) by income group	
			High (DALY _h)	Low+Middle (DALY _l)
Acute hepatitis B	Viral	0	82	1969
Ascariasis	Parasitic	1	0	681
Chagas disease	Parasitic	1	8	170
Cysticercosis	Parasitic	1	43	1063
Dengue	Viral	1	7	1742
Food-borne trematodes	Parasitic	1	33	861
Gonorrhoea	Bacterial	0	5	68
HIV/AIDS	Viral	0	2419	37749
Hookworm disease	Parasitic	1	0	534
Leishmaniasis	Parasitic	1	13	857
Leprosy	Bacterial	1	0	28
Lymphatic filariasis	Parasitic	1	3	1320
Malaria	Parasitic	0	3	52046
Onchocerciasis	Parasitic	1	0	1307
Rabies	Viral	1	9	2588
Schistosomiasis	Parasitic	1	4	1851
Syphilis	Bacterial	0	30	6111
Trachoma	Bacterial	1	0	134
Trichomoniasis	Parasitic	0	36	236
Trichuriasis	Parasitic	1	0	188
Trypanosomiasis	Parasitic	1	0	69
Tuberculosis	Bacterial	0	716	59867

The harm from communicable diseases can be measured as disability-adjusted life years lost (DALYs).

Recall



Optimal Funding Mechanisms



Different funding mechanisms...

- ... for malaria, TB and HIV/AIDS
- They cause similar level of pain and are often considered as similar by funding agencies
- Potential commercial profit quite different
 - Different prevalence in high-income countries

A Small Digression:

Mountains beyond Mountains by Tracy Kidder

- Paul Farmer (1959-2022)
 - Global Health
 - Duke and Harvard
 - Partners in Health
- Market Failure
 - TB drug pricing
 - Jim Yong Kim
 - Partners in Health, Dartmouth College, World Bank



Other Problems in Drug Development

- We use optimization models for other incentive issues in drug development
- Conditional Approval (FDA policy)

Conditional and Lenient Drug Approvals (2025) by Xu, Lopomo, Ridley, Sun

- Drugs for rare diseases (low market profit)
- Timing! (high profit only in patent protection period)
- FDA's "pre-approval program"
- Allow drug on market after Phase II while doing Phase III trial?
- Boost profit to incentivize investments into clinical trial
- Commit to lenient final approval threshold as well?

Mechanism Design

Auctions as Resource Allocation Mechanisms

- Resource Allocation (e.g., FCC's bandwidth auction)
- Procurement auctions: sellers are bidders
 - Vendors in the private sector
 - Builders for public contracts
- Key component: **private information**
 - Each bidder knows their own valuation of obtaining the resource, or production cost in procurement settings
- Potential objectives for resource allocation decisions: maximize
 - auctioneer's revenue, or
 - efficiency (sell to the bidder with the highest valuation/buy from the cheapest source)

General Mechanisms

- Posting a price
- Negotiation
- Various auctions
 - Sealed bid (first- and second- price auctions)
 - Ascending bid
 - Descending bid
 - ...
- Strategic behavior: e.g., bid may not be the same as valuation
- Which one maximizes the designer's objective?

Revenue Equivalence



- Revenue Equivalence Theorem: Myerson (1981) generalized Vickrey (1961)
- ~ Any mechanisms that yield the same equilibrium allocation outcome generate the same expected revenue.
- More importantly, Myerson formulated mechanism design as an optimization model.

Direct Mechanism and Optimization

- Revelation Principle:

Without loss of generality, focus on “**direct mechanisms**,” which induces truth-telling (e.g., bidding true value), and look for the *optimal* **direct mechanism**.

- **Optimal** mechanism design on a conceptual level

- **Objective:** Auctioneer’s expected revenue
- **Decision variables:** given a sequence of bids, *how to allocate* the item, and *how much to charge*
- **Constraints:**
 - bidder has incentive to bid the true valuation (incentive compatibility)
 $Utility(bidding\ true\ value) \geq Utility(bidding\ aother\ value)$
 - bidder willing to participate (individual rationality)
 $Utility(bidding\ true\ value) \geq 0$

Impact of Optimization Formulation

- Based on this optimization formulation, Myerson showed “second-price auction *with a reservation price*” is **optimal** when allocating one item among multiple bidders:
 - Maximizes seller’s revenue among all possible mechanisms
 - Not “efficient” due to reservation price – (may not allocate)
- Variations of this optimization formulation is behind designing optimal mechanisms for diverse settings
 - Multiple items? Dynamic arrivals? ...
- We considered a *concrete* setting in global health. But before that, let us *generalize* even more...

Adverse Selection and Moral Hazard

Asymmetric information

- Private information is not limited to bidder's valuation in an auction
- “Market for lemons” (Akerlof 1970)
 - Seller knows quality of used car
 - Buyer does not
 - Given uncertainty, buyers do not want to pay as if knowing car is of high quality
 - High quality sellers leave market
 - Only low-quality cars left, **market failure**
- Implication to insurance market:
 - Healthy people do not want to buy insurance

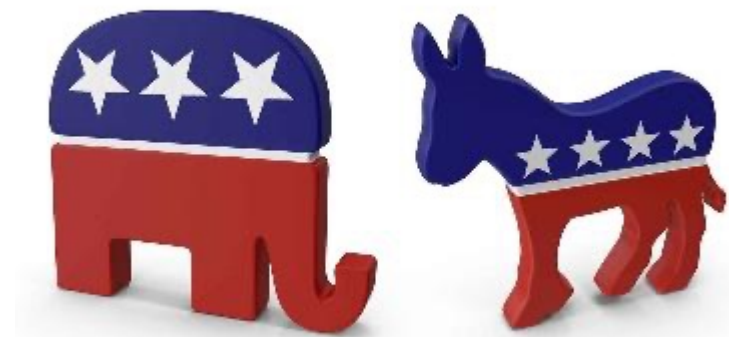


More on Insurance Market

- Private information: health conditions
- People with poorer health more willing to buy insurance – **Adverse Selection**
- Private actions: Smoke? Drive fast? Exercise?
- How to induce good behavior – **Moral Hazard**
- Two types of private information
 - **Adverse Selection** refers to agents know more than the designer, but does not “control” the uncertainty
 - **Moral Hazard** refers to agents take actions that not observable by the designer

Adverse Selection v.s. Moral Hazard

- Good incentive management/policy takes both adverse selection and moral hazard into consideration
 - Induce healthy people to buy insurance
 - Induce people to quit smoking, eating well, exercise
- Traditional political left/right division through the lens of asymmetric information
 - Different outcomes due to
 - Luck?
 - Effort?
- In business management?



Optimization for Incentive Management

- Asymmetric information
 - Adverse selection (mechanism design)
 - Moral hazard (contract design)
 - Designer has more information (information design)
- Operational levers
 - Monitoring/inspection
 - Adding delay to screen patient versus impatient customers in congested service systems
- Dynamic settings
 - Resource allocation without money
 - Learning and Statistics

Other Applications of Optimization

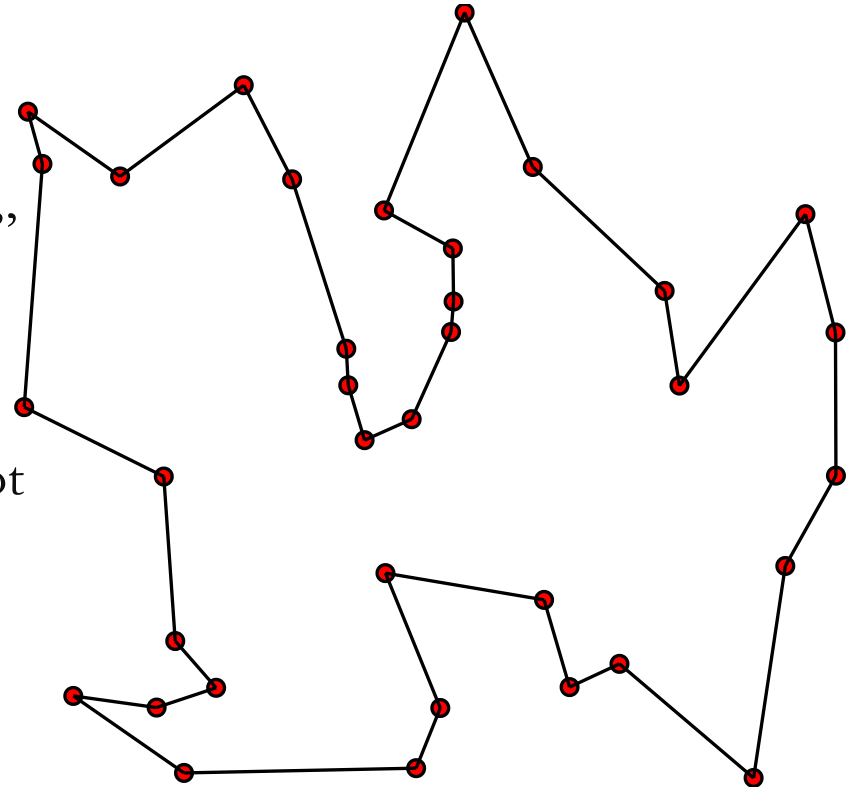
Delivery Routing Problem

- Summer of 2020
- Pandemic, local restaurants, Chapel Hill high school student drivers
- Picking up from a restaurant and delivering to multiple homes
- What is the best route for each driver?
- Formulate the optimization problem as Traveling Salesman Problem (TSP)
- Solve optimization using Google OR Tools:
<https://developers.google.com/optimization/install>
- Solution coded in Python by motivated middle school students
- Open source on GitHub

Traveling Salesman Problem (TSP)

“Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?”

- Theoretical computer science and complexity theory
- Algorithm design (enumeration does not work)
- Applications: logistics, microchip manufacturing, DNA sequencing, astronomy...
- Benchmark problem for quantum computing



Converting Delivery Routing to TSP

- Is our problem a TSP?
 - Restaurant, delivery addresses and driver's home as “nodes”
 - Pair-wise distance/time from Google Distance Matrix api or OpenStreetMap api
- Want to have a tour starting from restaurant and ending at driver's home
- What if the TSP tour puts restaurant in between delivery addresses?
 - Revise travel distance/time from any delivery address to restaurant to be extremely large, and from driver's home to restaurant 0

Web based interface

- <http://routing.colab.duke.edu/delivery/>
- Open source: <https://github.com/autorouting/main>

The screenshot shows the 'Autorouting' web interface. At the top, there's a 'Select Language' dropdown and a 'Powered by Google Translate' note. Below this is a 'Introduction Video' button and a link to 'Or read instructions'. A note states: 'To avoid confusion, please specify the city and state of every address.' The form has three main sections: 'Starting address' with a text input containing 'Jade Palace, Chapel Hill, NC'; 'Intermediate addresses (one address per line, fifty lines max)' with a text area containing a list of addresses: '306 Pleasant Dr, Cambo, NC 27510', '206 Northwood Dr, Chapel Hill, NC 27514', '112 Saddle Ridge Rd, Chapel Hill, NC 27516', '110 Unwood Dr, Chapel Hill, NC 27516', and '302 Pleasant Dr, Cambo, NC 27510'; and 'Ending address' with a text input containing '110 Cabernet Drive, Chapel Hill, NC'. There's also a 'Your Email (Optional)' field with 'me@example.com' and a checkbox for 'Speed up calculation (this will reduce the quality of the outcome)' which is checked. A 'Launch program' button is at the bottom.

